



Lesson 6.5 — Properties of Logarithms

Big idea: Logarithms turn multiplication into addition, division into subtraction, and exponents into multipliers. Three short rules cover almost everything.

Quick review

Product rule: $\log_b(MN) = \log_b M + \log_b N$.

Quotient rule: $\log_b(M/N) = \log_b M - \log_b N$.

Power rule: $\log_b(M^p) = p \cdot \log_b M$.

Change of base: $\log_b(x) = \log(x)/\log(b) = \ln(x)/\ln(b)$.

Don't do: $\log(M + N) \neq \log M + \log N$. Logs are linear under MULTIPLICATION, not addition.

Worked example

Worked example. Expand $\log_5(25x^3/\sqrt{y})$.

$$= \log_5(25) + \log_5(x^3) - \log_5(y^{1/2}) = 2 + 3 \log_5 x - (1/2) \log_5 y.$$

Warm-ups

1. Expand $\log(xy)$.
2. Expand $\log(x/3)$.
3. Expand $\log(x^2)$.

Practice

1. Expand $\log_2(8x^3)$.
2. Condense $2 \log x - \log y + 3 \log z$ into one log.
3. Use change of base to evaluate $\log_5(40)$ (round to 4 decimals).

Challenge

1. Solve $\log_2(x) + \log_2(x - 2) = 3$ for x .
2. Show that $\log_b(M^p) = p \cdot \log_b M$ using the definition of log and exponent rules.



Answer key — Lesson 6.5

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $\log x + \log y$.
2. $\log x - \log 3$.
3. $2 \log x$.

Practice

1. $3 + 3 \log_2 x$.
2. $\log(x^2 z^3 / y)$.
3. $\ln(40) / \ln(5) \approx 3.6889 / 1.6094 \approx 2.2920$.

Challenge

1. Combine: $\log_2[x(x-2)] = 3 \Rightarrow x(x-2) = 8 \Rightarrow x^2 - 2x - 8 = 0 \Rightarrow (x-4)(x+2) = 0$. $x = 4$ (reject -2 because log domain).
2. Let $\log_b M = y$ so $M = b^y$. Then $M^p = (b^y)^p = b^{py}$, so $\log_b(M^p) = py = p \log_b M$. ✓